MICROSOFT DEFENDER SMART SCREEN WITH GROUP POLICY CONFIGURATION

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Configuring Microsoft Defender SmartScreen with Group Policy

You are working on securing your Windows Servers, and as part of the effort, you want to add protections against phishing and malware from websites, applications, and downloaded files. In this hands-on lab, you are going to create a Group Policy for a domain-joined virtual machine. The policy will enforce enabling SmartScreen on any domain-joined machine.

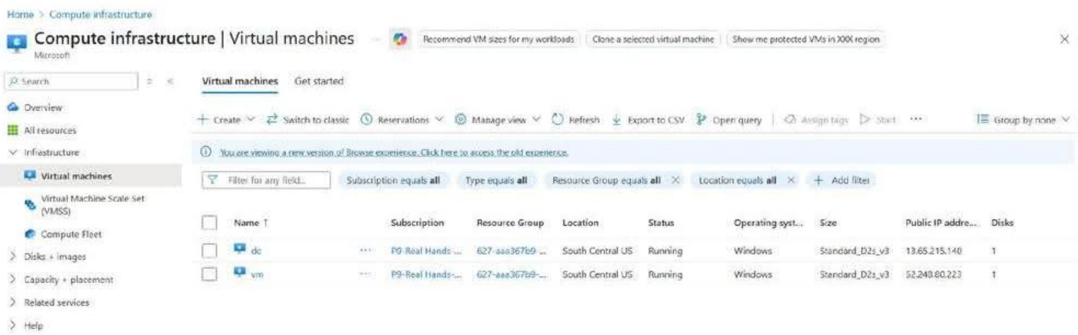
Solution

Log in to the Azure portal. Connect to the `dc` and `vm` virtual machines via remote desktop (either through the Remote Desktop client available on Windows machines or through the Microsoft Remote Desktop application available for Mac machines) using the public IP address,

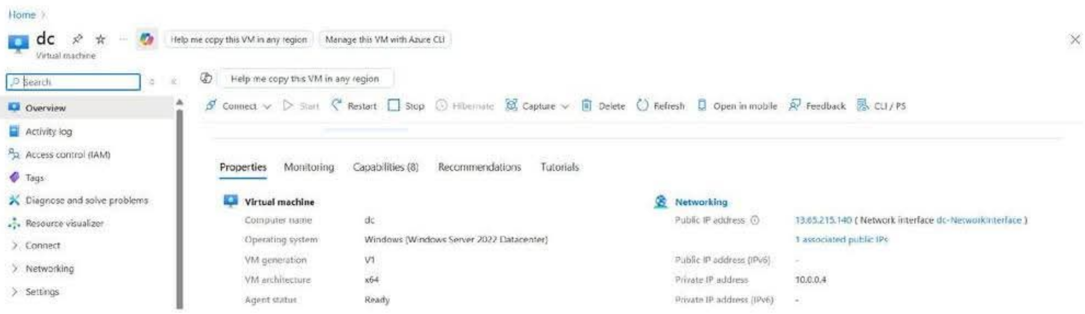
Configure the Active Directory Domain Services Environment

Confirm That the Private IP Address of the Domain Controller VM Is Set to Static

1. In the Azure portal, from the list of resources, click the **dc** virtual machine to open the domain controller VM.

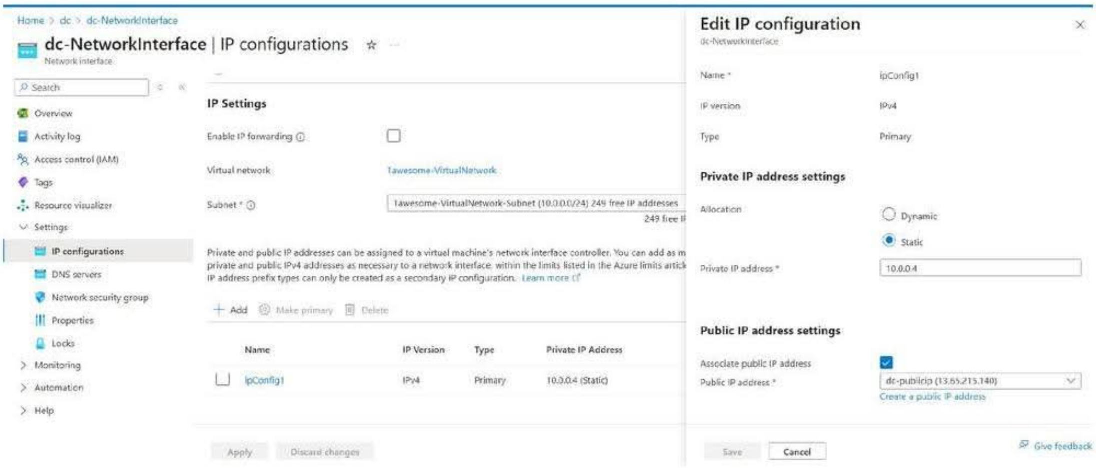


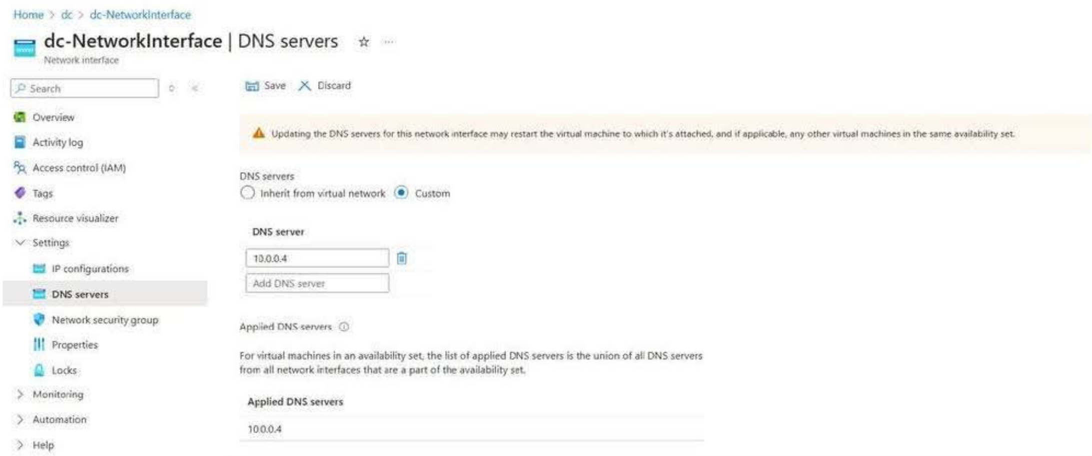
2. In the navigation menu on the left, under **Settings**, click **Networking**.
3. Click the link for the **dc-NetworkInterface**.



4. In the menu on the left, under **Settings**, click **IP configurations**.

5. On the **IP configurations** page for the domain controller, note the IP address in the **Private IP address** field — it should be `10.0.0.4` — and confirm that it also displays **(Static)**.



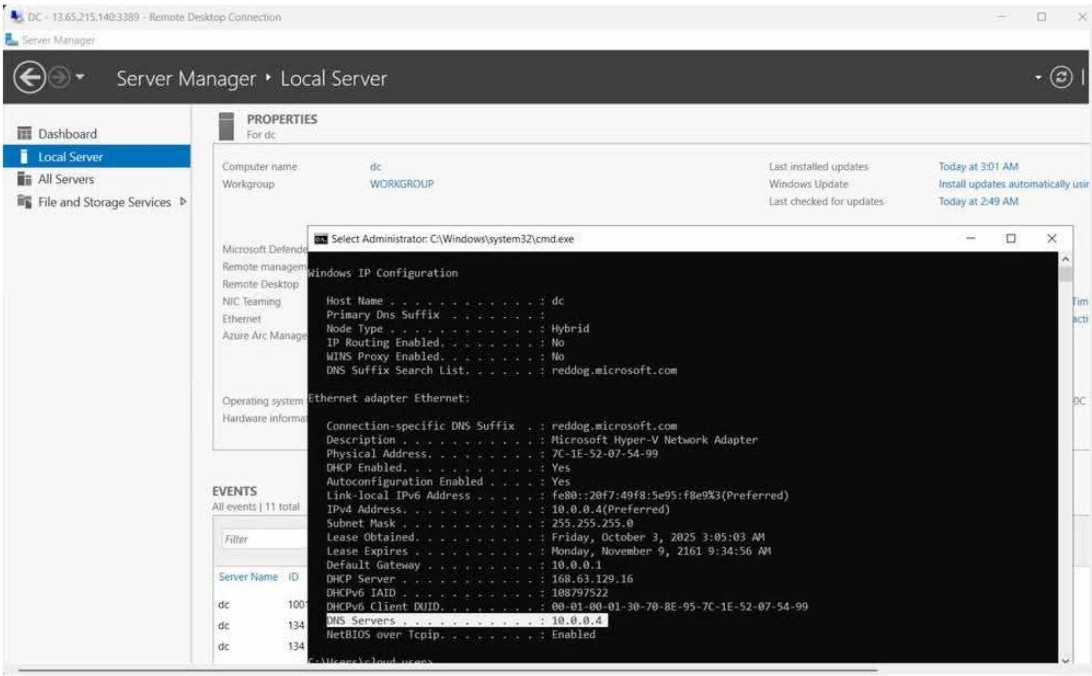


> Note: If for some reason it is not set to Static, click the configuration and toggle the Assignment field to Static and set the IP address.

Install Active Directory Domain Services on the Domain Controller VM

1. In the breadcrumb trail at the top of the screen, click **dc** to navigate back to the virtual machine.
2. In the menu on the left, click **Overview**.
3. From the menu at the top, click the **Connect** down-arrow and select **RDP**.
4. Connect to the `dc` virtual machine via remote desktop, using the public IP address and credentials provided on the lab page.

   > Note: The Server Manager application will open automatically when you connect to the VM. Leave it open, as you will use this later on in the lab. If for some reason it isn't launched, you can do so by clicking the Start menu and selecting Server Manager.



5. On the `dc` virtual machine, click the **Start** menu icon.
6. From the **Start** menu, open **Windows PowerShell ISE**.
7. Click the **Copy raw contents** button.

```
# Install AD DS and configure forest

$ProgressPreference = "SilentlyContinue"
$WarningPreference = "SilentlyContinue"
Install-WindowsFeature "AD-Domain-Services" -IncludeManagementTools | Out-Null
$pw = ConvertTo-SecureString "p@55w0rd" -AsPlainText -Force
Install-ADDSForest -DomainName "awesome.com" -SafeModeAdministratorPassword $pw
-DomainNetBIOSName 'CORP' -InstallDns -Force -NoRebootOnCompletion
```

If you are not comfortable with ps when setup dc just do as usual using GUI mode, no worries...!

8. Navigate back to PowerShell and paste the copied script in the top pane.

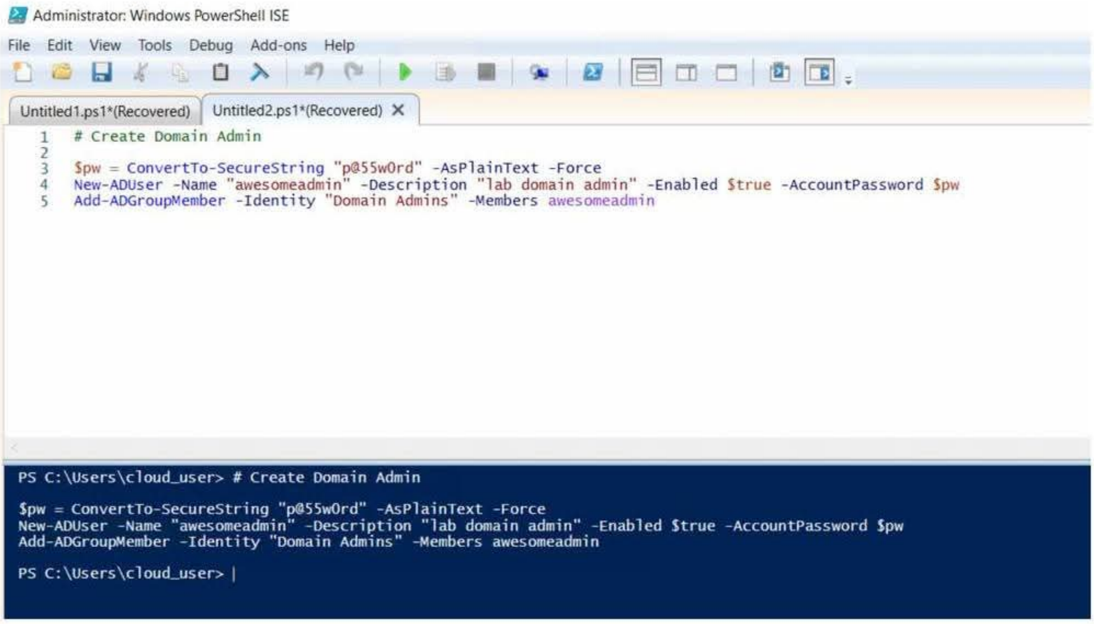
> Note: If the script pane does not display by default, select View from the menu at the top of the PowerShell window and toggle on the Show Script Pane option.

9. Review the actions provided in the `installadds` script: installing Active Directory Domain Services, including the management tools; setting the password; and then installing/configuring the AD DS forest with a domain name of **awesome.com**.
10. From the menu at the top, click the **Run Script** button (or press **F5**).
11. Once the script has completed, close PowerShell without saving the file.
12. Navigate back to the Azure portal and, from the menu at the top, click **Restart** and select **Yes**.

Create the Domain Admin User on the Domain Controller VM

1. Once the `dc` virtual machine has restarted, connect to it via remote desktop (using the public IP address and credentials provided on the lab page) and launch PowerShell ISE again.
2. Click the **Copy raw contents** button.

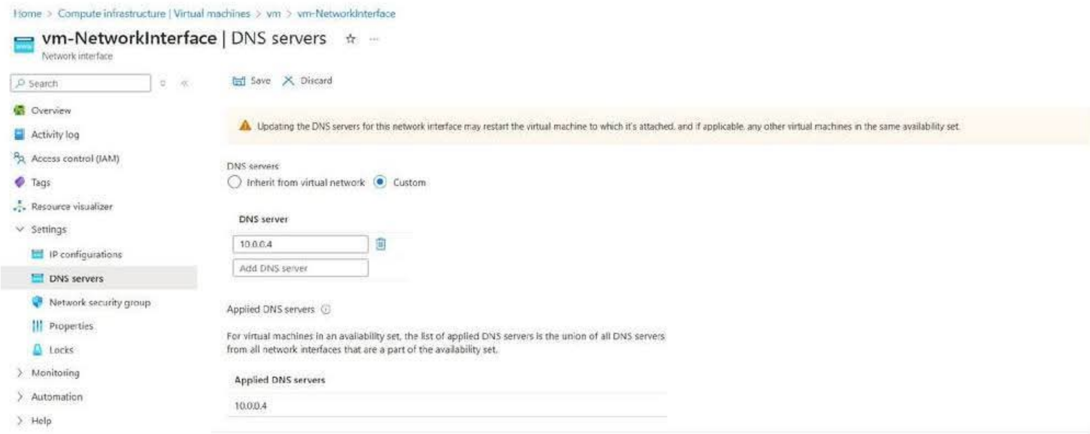
```
# Create Domain Admin

$pw = ConvertTo-SecureString "p@55w0rd" -AsPlainText -Force
New-ADUser -Name "awesomeadmin" -Description "lab domain admin" -Enabled $true -AccountPassword $pw
Add-ADGroupMember -Identity "Domain Admins" -Members awesomeadmin
```



3. Navigate back to PowerShell and paste the copied script in the top pane.
4. Review the actions provided in the `domainadmin` script: adding a Domain Admin user named **awesomeadmin** and adding that user to the **Domain Admins** group.
5. From the menu at the top, click the **Run Script** button (or press **F5**).

Confirm That the DNS for the VM That Will Be Joining the Domain Is Set to the Private IP of the Domain Controller VM

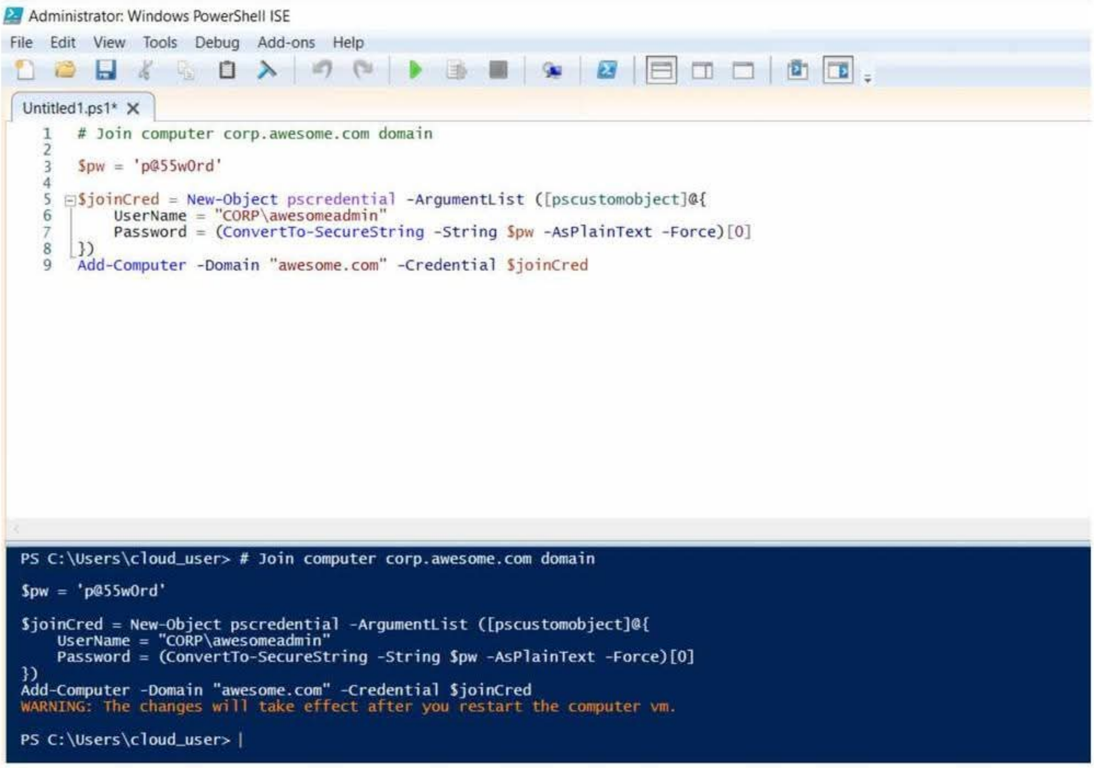
1. In the Azure portal, in the breadcrumb trail at the top of the screen, click the lab title link to navigate back to the resource group.
2. From the list of resources, click the **vm** virtual machine to open the VM.
3. In the navigation menu on the left, under **Settings**, click **Networking**.
4. Click the link for the **vm-NetworkInterface**.
5. In the menu on the left, under **Settings**, click **DNS servers**.
6. Verify that the **Custom** option is selected and the DNS server has been set to the private IP address of the domain controller VM, which is `10.0.0.4`.



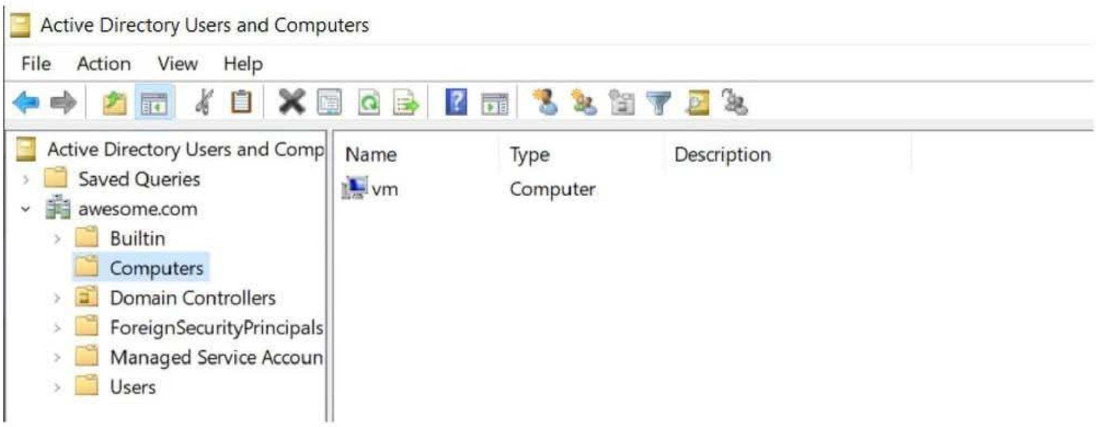
Join the VM to the Domain

1. In the breadcrumb trail at the top of the screen, click **vm** to navigate back to the virtual machine.
2. In the menu on the left, click **Overview**.
3. From the menu at the top, click the **Connect** down-arrow and select **RDP**.
4. Connect to the `vm` virtual machine via remote desktop, using the public IP address and credentials provided on the lab page.
5. On the `vm` virtual machine, click the **Start** menu icon.
6. From the **Start** menu, open **Windows PowerShell ISE**.
7. Click the **Copy raw contents** button.

```
# Join computer corp.awesome.com domain

$pw = 'p@55w0rd'

$joinCred = New-Object pscredential -ArgumentList ([pscustomobject]@{
   UserName = "CORP\awesomeadmin"
   Password = (ConvertTo-SecureString -String $pw -AsPlainText -Force)[0]
})
Add-Computer -Domain "awesome.com" -Credential $joinCred
```

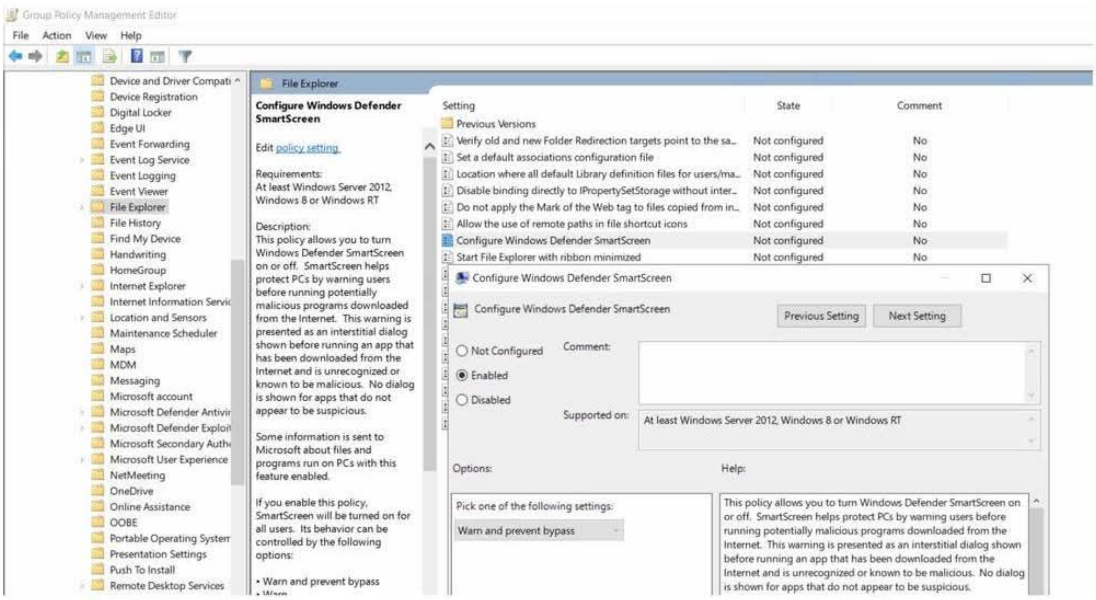


8. Navigate back to PowerShell and paste the copied script in the top pane.
9. Review the actions provided in the `domainjoin` script: adding this computer to the **awesome.com** domain, using the credentials saved in the `$joinCred` parameter.
10. From the menu at the top, click the **Run Script** button (or press **F5**).
11. Once the script has completed, close PowerShell without saving the file.
12. Navigate back to the Azure portal and, from the menu at the top, click **Restart** and select **Yes**.
13. Once the `vm` virtual machine has restarted, connect to it via remote desktop (using the public IP address and credentials provided on the lab page).
14. In **Server Manager**, from the menu on the left, click **Local Server**.
15. In the **PROPERTIES** section, confirm that the **vm** virtual machine is joined to the **awesome.com** domain.



Configure Group Policy for SmartScreen

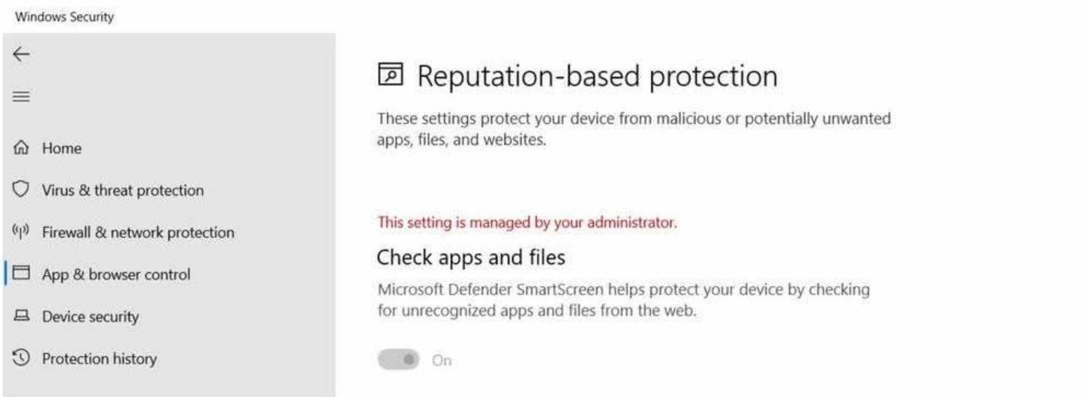
1. On the `dc` virtual machine, in **Server Manager**, click **Tools** at the top-right of the screen and then select **Group Policy Management** from the menu.
2. In the **Group Policy Management** window, in the left navigation menu, expand **Forest: awesome.com**, and then expand **Domains**.
3. Expand and select **awesome.com**.
4. From the menu at the top, click **Action** and then select **Create a GPO in this domain, and Link it here....**
5. In the **New GPO** dialog box, in the **Name** field, enter *smartscreen* and click **OK**.
6. Once the **smartscreen** policy appears in the left navigation menu, select it.
7. In the **Group Policy Management Console** dialog box, click **OK**.
8. From the menu at the top, click **Action** and then select **Edit**.
9. In the **Group Policy Management Editor** window, in the left navigation menu, under **Computer Configuration**, expand **Policies**, **Administrative Templates**, and **Windows Components** and then select **File Explorer**.
10. In the right pane, select **Configure Windows Defender SmartScreen**.
11. In the description that appears for the selected setting, click the **Edit policy setting** link.
12. In the dialog box, select **Enabled**.
13. Click **Apply**, and then click **OK**.



Confirm SmartScreen Policy Has Been Applied

1. On the `vm` virtual machine, in the **Search** bar in the taskbar, type *cmd* and select **Command Prompt**.
2. In the command line, type `gpupdate /force` and press **Enter**.
3. Once the policy has been updated, from the **Start** menu, open **Windows Security**.

4. In the **Windows Security** window, in the left navigation menu, select **App & browser control**.
5. Under **Reputation-based protection**, select the **Reputation-based protection settings** link.
6. Confirm that the **Check apps and files** option has been turned **On**, but is greyed out (i.e. cannot be modified) because the setting is managed by your administrator — in other words, the Group Policy, which was configured to enable Microsoft Defender SmartScreen for any domain-joined virtual machine, was applied successfully to this VM.



Thank You...!